

5. (a) Define cardiac output. Describe the various factors that affect the regulation of cardiac output.
- (b) Draw a neat labelled diagram of cardiac cycle showing all the events associated with one heartbeat. (6,6)
6. (a) Describe the three phases of digestion.
- (b) Explain the factors involved in regulation of blood pressure. (6,6)
7. Write short notes on **Any Three** of the following : (3×4=12)
- (a) Pulmonary volume and capacities
- (b) Glomerular filtration
- (c) ECG
- (d) Carbon monoxide poisoning
- (e) Formed elements of blood

[This question paper contains 4 printed pages.]

03.06.24(14)
Your Roll No.....

Sr. No. of Question Paper : 3010

H

Unique Paper Code : 32231402

Name of the Paper : Animal Physiology Life Sustaining Systems

Name of the Course : B.Sc. (Hons.) Zoology

Semester : IV, LOCF

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

- Write your Roll No. on the top immediately on receipt of this question paper.
- Attempt any **FIVE QUESTIONS** in all.
- Question no. 1 is **COMPULSORY**.
- Draw diagrams wherever necessary.

1. (a) Define the following terms : (5)
- Hematocrit
 - Respiratory acidosis
 - Haustral churning
 - Renal threshold
 - Ectopic focus

(b) Distinguish between the following : (10)

- (i) Bohr and Haldane effect
- (ii) PCT and DCT
- (iii) Pulmonary ventilation and pulmonary respiration
- (iv) Hemopoiesis and hemostasis
- (v) Peristalsis and segmentation

(c) Expand the following abbreviations : (2)

- (i) PCV
- (ii) EDV
- (iii) SARS
- (iv) CCK

(d) Fill in the blanks : (4)

- (i) Kupffer cells are found in _____ and play a role in _____.
- (ii) The renal corpuscle consists of the _____ and _____.

(e) State the *location* and *function* of the following (Any FOUR) : (4)

- (i) Brunner's glands
- (ii) Septal cells

(iii) Apneustic area

(iv) JGA

(v) Crypts of Lieberkühn

(f) Give reason/s : (2)

(i) Tetany does not occur in the cardiac cells.

(ii) Digested lipids are mainly absorbed in the lacteals.

2. (a) Discuss the counter current mechanism and its role in producing concentrated urine.

(b) Explain the mechanism of GFR regulation in nephrons. (6,6)

3. (a) Discuss the basic characteristics of alveoli for gas exchange: Explain the role of various muscles involved in pulmonary ventilation.

(b) Briefly explain the factors affecting Oxygen-hemoglobin dissociation curve. (6,6)

4. (a) Draw and explain portal triad. Briefly discuss the functions of liver.

(b) How is HCl formed in the stomach? (8,4)